Multiple Choice:

1. Find x: $\begin{pmatrix} 1 & 2 & -3 \\ 4 & 1 & 3 \end{pmatrix} - x = \begin{pmatrix} 5 & 1 & 8 \\ -6 & 0 & 5 \end{pmatrix}$.

   A. $\begin{pmatrix} 4 & 1 & -11 \\ -8 & 1 & -2 \end{pmatrix}$
   B. $\begin{pmatrix} -4 & -1 & -11 \\ 8 & -1 & 2 \end{pmatrix}$
   C. $\begin{pmatrix} -5 & -2 & 24 \\ 12 & 0 & -15 \end{pmatrix}$
   D. $\begin{pmatrix} 6 & 3 & 5 \\ -4 & 1 & 8 \end{pmatrix}$

2. Find the adjoint matrix of $\begin{bmatrix} 1 & -4 \\ 5 & -3 \end{bmatrix}$.

   A. $\begin{bmatrix} 1 & 0 \\ 4 & 2 \end{bmatrix}$
   B. $\begin{bmatrix} 2 & -3 \\ 5 & 1 \end{bmatrix}$
   C. $\begin{bmatrix} -3 & -5 \\ 4 & 1 \end{bmatrix}$
   D. $\begin{bmatrix} 2 & 4 \\ 0 & 1 \end{bmatrix}$

3. One of the eigen vectors of the matrix $A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$ is _____.

   A. $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$
   B. $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
   C. $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$
   D. $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$

4. Which of the following equations has $-i$, i, and 0 as its only roots?

   A. $x^3 - 1 = 0$
   B. $x^3 - x = 0$
   C. $x^3 + 1 = 0$
   D. $x^3 + x = 0$

5. Determine the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$.

   A. 1
   B. 3
   C. 2
   D. 4

6. Laplace transform of the function $\sin \omega t$ is _______.

   A. $\dfrac{s}{s^2 + \omega^2}$
   B. $\dfrac{s}{s^2 - \omega^2}$
   C. $\dfrac{\omega}{s^2 + \omega^2}$
   D. $\dfrac{\omega}{s^2 - \omega^2}$

7. Calculate $\dfrac{8 + 5i}{1 - 2i}$.

   A. $\dfrac{2}{5} + \dfrac{21}{5}i$
   B. $-\dfrac{2}{5} - \dfrac{21}{5}i$
   C. $-\dfrac{2}{5} + \dfrac{21}{5}i$
   D. $-\dfrac{21}{5} + \dfrac{2}{5}i$

8. If $\begin{pmatrix} 6 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ x \end{pmatrix}$, then x = _______.

   A. 0
   B. 5
   C. 1
   D. 4

9. Find the z-transform of $x[n] = \delta[n-5]$.

   A. $z^{-1/5}$
   B. z^5
   C. $z^{1/5}$
   D. z^{-5}

10. If $A = \begin{pmatrix} -2 & 5 \\ 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 6 & 0 \\ 3 & -5 \end{pmatrix}$. Evaluate BA.

    A. $\begin{pmatrix} -12 & 30 \\ -11 & 0 \end{pmatrix}$
    B. $\begin{pmatrix} -12 & 20 \\ -11 & 0 \end{pmatrix}$
    C. $\begin{pmatrix} -12 & 30 \\ -11 & -29 \end{pmatrix}$
    D. $\begin{pmatrix} -12 & 30 \\ 11 & 0 \end{pmatrix}$

11. Evaluate $2\log_{(1+j)}\left(1 + j\sqrt{3}\right)$

    A. $1.442 - j0.246$
    B. $1.198 - j0.215$
    C. $0.151 + j0.341$
    D. $2.88 - j0.492$

12. If $i = \sqrt{-1}$, then $\sum_{j=0}^{10} (-i)^J =$ _____.

    A. $-i$
    B. -1
    C. i
    D. 1

13. What is the coefficient of x in the Maclaurin series for $e^{\sin x}$?

    A. 0
    B. -1
    C. 1
    D. 1/2!

14. Compute the 4th Taylor polynomial of the function $f(x) = \sin x$ centered at $a = \dfrac{\pi}{2}$.

    A. $1 - \dfrac{1}{2}\left(x - \dfrac{\pi}{2}\right)^2 + \dfrac{1}{24}\left(x - \dfrac{\pi}{2}\right)^4$
    B. $1 - \dfrac{1}{2}\left(x - \dfrac{\pi}{2}\right)^2 + \dfrac{1}{24}\left(x + \dfrac{\pi}{2}\right)^4$

C. $1-\frac{1}{2}\left(x-\frac{\pi}{2}\right)^2-\frac{1}{24}\left(x-\frac{\pi}{2}\right)^4$

D. $1-\frac{1}{2}\left(x+\frac{\pi}{2}\right)^2+\frac{1}{24}\left(x-\frac{\pi}{2}\right)^4$

15. Determine the inverse matrix of $\begin{pmatrix} 1 & 2 \\ 5 & 9 \end{pmatrix}$.

A. $\begin{pmatrix} -9 & 5 \\ 2 & -1 \end{pmatrix}$

B. $\begin{pmatrix} 9 & 5 \\ 2 & 1 \end{pmatrix}$

C. $\begin{pmatrix} 2 & 5 \\ 9 & 1 \end{pmatrix}$

D. $\begin{pmatrix} -9 & -5 \\ 2 & 1 \end{pmatrix}$

16. Simplify: $i^{29}+i^{21}+i$

A. 3i
B. 1- i
C. 1 + i
D. 2i

17. Write in the form a + bi the expression $i^{3217} - i^{427} + i^{18}$

A. 2i + 1
B. – i + 1
C. 2i – 1
D. 1 + i

18. The Taylor Series of the expansion of 3sin x + 2cos x is:

A. $2+3x-x^2-\frac{1}{2}x^3+...$

B. $2+3x+x^2+\frac{1}{2}x^3+...$

C. $2-3x+x^2-\frac{1}{2}x^3+...$

D. $2-3x-x^2+\frac{1}{2}x^3+...$

19. For $A=\begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, the determinant of A^TA^{-1} is:

A. $\sec^2 x$
B. 1
C. cos 4x
D. 0

20. The value of x for which the matrix A $=\begin{bmatrix} 3 & 2 & 4 \\ 9 & 7 & 13 \\ -6 & -4 & -9+x \end{bmatrix}$ has zero as an eigenvalue is:

A. 1
B. – 1
C. 2
D. – 2

21. $\frac{1+2i}{1-i}$ lies in ______ quadrant.

A. first
B. second
C. third
D. fourth

22. What is the reciprocal of $3+\sqrt{7}i$?

A. $\frac{5}{16}+\frac{\sqrt{7}i}{16}$

B. $\frac{5}{16}-\frac{\sqrt{7}i}{16}$

C. $\frac{3}{16}+\frac{\sqrt{7}i}{16}$

D. $\frac{3}{16}-\frac{\sqrt{7}i}{16}$

23. What is the conjugate of $\frac{\sqrt{5+12i}+\sqrt{5-12i}}{\sqrt{5+12i}-\sqrt{5-12i}}$?

A. $1-\frac{3}{2}i$

B. $1+\frac{3}{2}i$

C. $0-\frac{3}{2}i$

D. $0+\frac{3}{2}i$

24. What is the polar form of the complex number $\left(i^{25}\right)^3$?

A. $\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}$

B. $\cos\frac{\pi}{2}-i\sin\frac{\pi}{2}$

C. $\cos\frac{\pi}{4}-i\sin\frac{\pi}{4}$

D. $\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}$

25. What is the inverse Laplace transform of $\frac{1}{\left(s^2+s\right)}$?

A. $1+e^t$
B. $1-e^{-t}$
C. $1-e^t$
D. $1+e^{-t}$

END